



Basic Components of a Drone

Understanding the basic components of a drone is essential in drone forensics because **each component may store physical or digital evidence.** A forensic examiner must analyze both hardware and software elements to reconstruct drone activity, establish operational timelines, and identify the operator. This comprehensive knowledge forms the foundation for effective digital forensic investigations involving unmanned aerial systems.

Hardware Components: Physical Evidence Sources

Hardware components are the physical parts of a drone that enable it to fly, navigate, capture data, and communicate. Each component serves as a potential evidence source in forensic investigations, providing critical information about drone operations, flight history, and operator behaviour.



Frame / Body

The **airframe is the physical structure holding all internal components**. Materials include **plastic** (consumer drones), **carbon fibre** (professional), or **aluminium composites**. Damage patterns reveal crash analysis data, whilst serial numbers identify manufacturer and model. Structural modifications may indicate illegal or custom-built drones.



Motors & Propellers

Brushless DC motors rotate propellers to generate lift and thrust. Propellers come in clockwise (CW) and counter-clockwise (CCW) types. Broken propellers indicate collision direction, motor overheating marks suggest overloading, and propeller mismatch may reveal tampering or replacement.



Li-Po Battery

Lithium-Polymer batteries provide lightweight, high-energy output. They store charge cycles, voltage levels, and temperature logs. Battery data estimates flight time, serial numbers link batteries to specific drones, and swollen batteries indicate improper usage or overheating during operations.



Flight Controller

The central processing unit receives **sensor and GPS data, processes pilot commands, and controls motor speed**. It stabilises the drone, executes flight commands, maintains altitude and direction, and stores flight logs containing telemetry data crucial for reconstructing entire flight paths.

Critical Navigation & Sensing Hardware

GPS Module – Location Tracking

The **Global Positioning System** module determines the drone's exact geographical position, enabling navigation, Return-to-Home (RTH) functionality, and geo-fencing capabilities.

Forensic Significance:

- Identifies precise take-off and landing locations
- Confirms entry into restricted or no-fly zones
- Links drone activity to crime scene locations
- Provides coordinate data for mapping flight paths
- Validates timestamps against location data

Camera & Gimbal – Media Capture

The **camera captures images and videos** whilst the **gimbal (2-axis or 3-axis) stabilises the camera for smooth footage**.

Photos and videos act as direct digital evidence, with EXIF metadata revealing date, time, GPS location, and camera settings. Gimbal angle shows direction of surveillance and target focus.

Sensors – Movement & Environment

Sensors provide real-time data about drone movement and environmental conditions, essential for flight stability and forensic reconstruction.

Gyroscope: Maintains **orientation and balance** throughout flight operations, detecting rotational changes in three-dimensional space.

Accelerometer: Measures **acceleration, deceleration, and speed changes**, providing crucial data for impact analysis and flight behaviour patterns.

Barometer: Measures **altitude using air pressure, offering precise height data that complements GPS** altitude readings.

Sensor data helps detect sudden drops, collisions, validates flight behaviour, and supports accurate event timeline reconstruction in forensic investigations.

Software Components: Digital Evidence Repositories

Software components control drone operations, store digital logs, and enable communication between drone and operator. These represent rich sources of digital evidence that forensic examiners must thoroughly analyze to reconstruct operator intent, flight history, and operational patterns.

1

Firmware – Embedded Control Software

Firmware is **low-level embedded software installed inside the drone that directly controls hardware components** including motor control, sensor calibration, and communication management. Firmware version reveals update history and maintenance patterns, whilst modified firmware may indicate unauthorised or illegal drone usage. Logs stored in firmware provide detailed operational data for forensic analysis, including system errors, calibration events, and hardware interactions.

2

Mobile Applications

Mobile applications such as **DJI Fly, DJI GO, and Litchi** act as the user interface between operator and drone. They provide live video feed, flight control, data synchronisation, and cloud backup capabilities. Apps store flight logs, cached media files, and user account details that link the drone to a specific individual. Application data provides extended flight history beyond what's stored on the drone itself, including deleted or overwritten flights.

3

Ground Control Software (GCS)

Ground Control Software is used on **computers or tablets for advanced drone operations**, including mission planning, waypoint navigation, autonomous flight control, and real-time telemetry monitoring. Mission files reveal pre-planned routes and operational intent, confirming whether operations were autonomous or manual. **GCS provides crucial evidence of planning, intent, and sophisticated operational knowledge by the operator.**

- ❏ **Forensic Best Practice:** Always examine both hardware and software components in parallel. Cross-referencing data from multiple sources validates findings and reveals inconsistencies that may indicate tampering, data manipulation, or attempts to conceal operational history.

The Drone Anatomy Analogy

"A drone is like a human body – **Frame** is the skeleton, **Motors** are muscles, **Sensors** are sense organs, **Flight controller** is the brain, and **battery** is the heart."

This analogy helps forensic examiners understand component interdependencies and evidence correlation. Just as medical examiners analyze bodily systems to determine cause and sequence of events, drone forensic investigators must examine component relationships to reconstruct flight operations, identify anomalies, and establish operator actions.



Frame = Skeleton

Provides structural support and protection for all internal systems



Motors = Muscles

Generate movement and enable physical actions through controlled power



Sensors = Sense Organs

Perceive environmental conditions and spatial awareness data



Flight Controller = Brain

Processes information and coordinates all system functions



Battery = Heart

Powers all systems and sustains operational life

Understanding this systems approach enables forensic examiners to develop comprehensive investigation strategies, identify evidence sources across multiple components, and reconstruct complete operational narratives from fragmented data sources.